



AI-Driven Load Forecasting & Anomaly Detection

for Power Grid Analytics: A Comparative Study Using
Machine Learning and Deep Learning

Md Rayhan Chowdhury¹

¹Dept. of Electrical & Computer Engineering, [Lamar University] mchowdhury15@lamar.edu

IEEE
PES General Meeting
Power Systems
2025

2.34%

Best MAPE
XGBoost

Motivation & Problem Statement

Integrating **AI data centers**, EV fleets, and large industrial loads is driving unprecedented volatility in modern power grids. Critical gaps limit existing work:

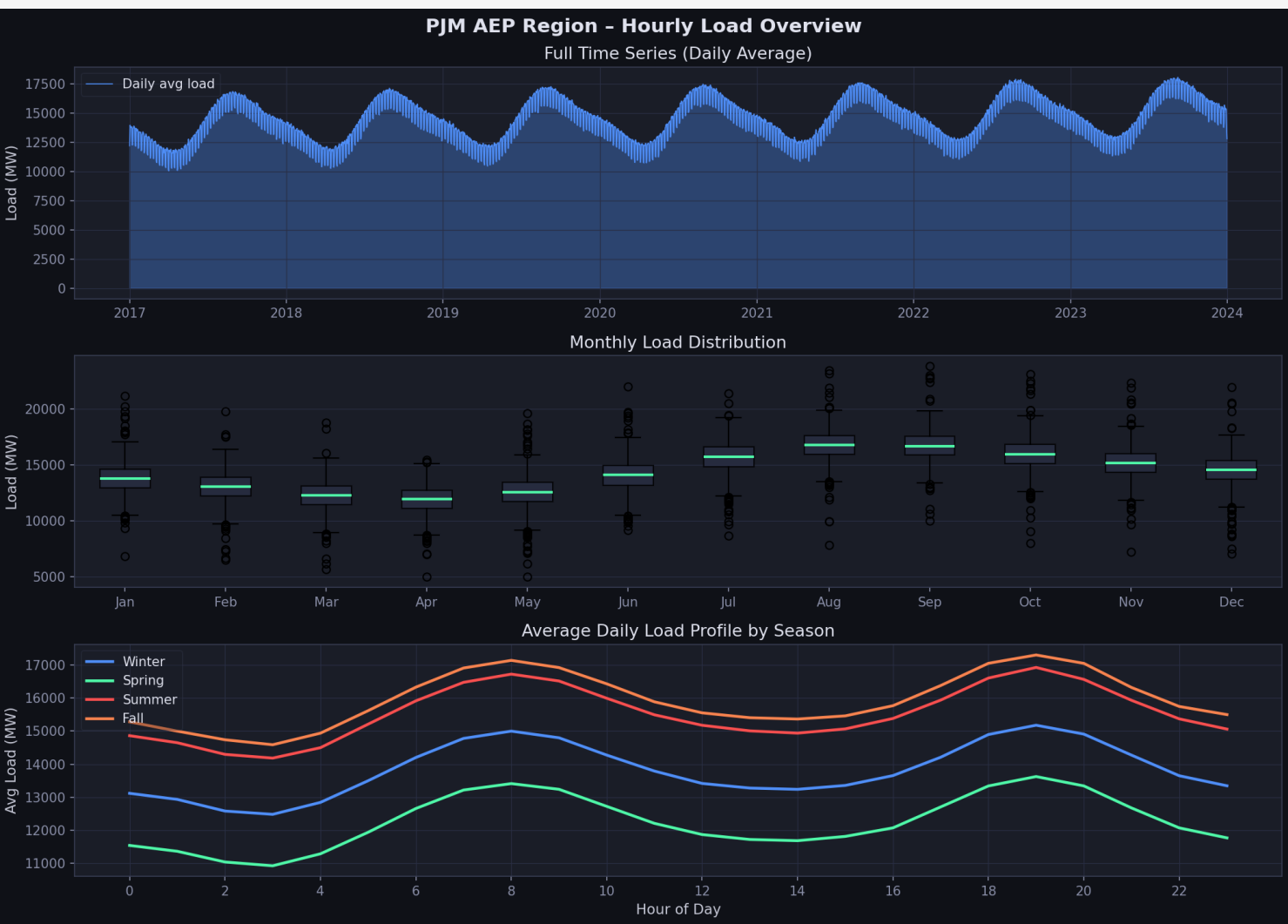
- **Forecasting accuracy** degrades under high-inertia, nonlinear seasonal demand patterns
- **AI data center ramps** create new load signatures unaccounted for in classical grid analytics frameworks

Research Questions

RQ1: Which unsupervised anomaly detector (Isolation Forest vs. LSTM Autoencoder) better identifies abnormal grid events without labeled data?

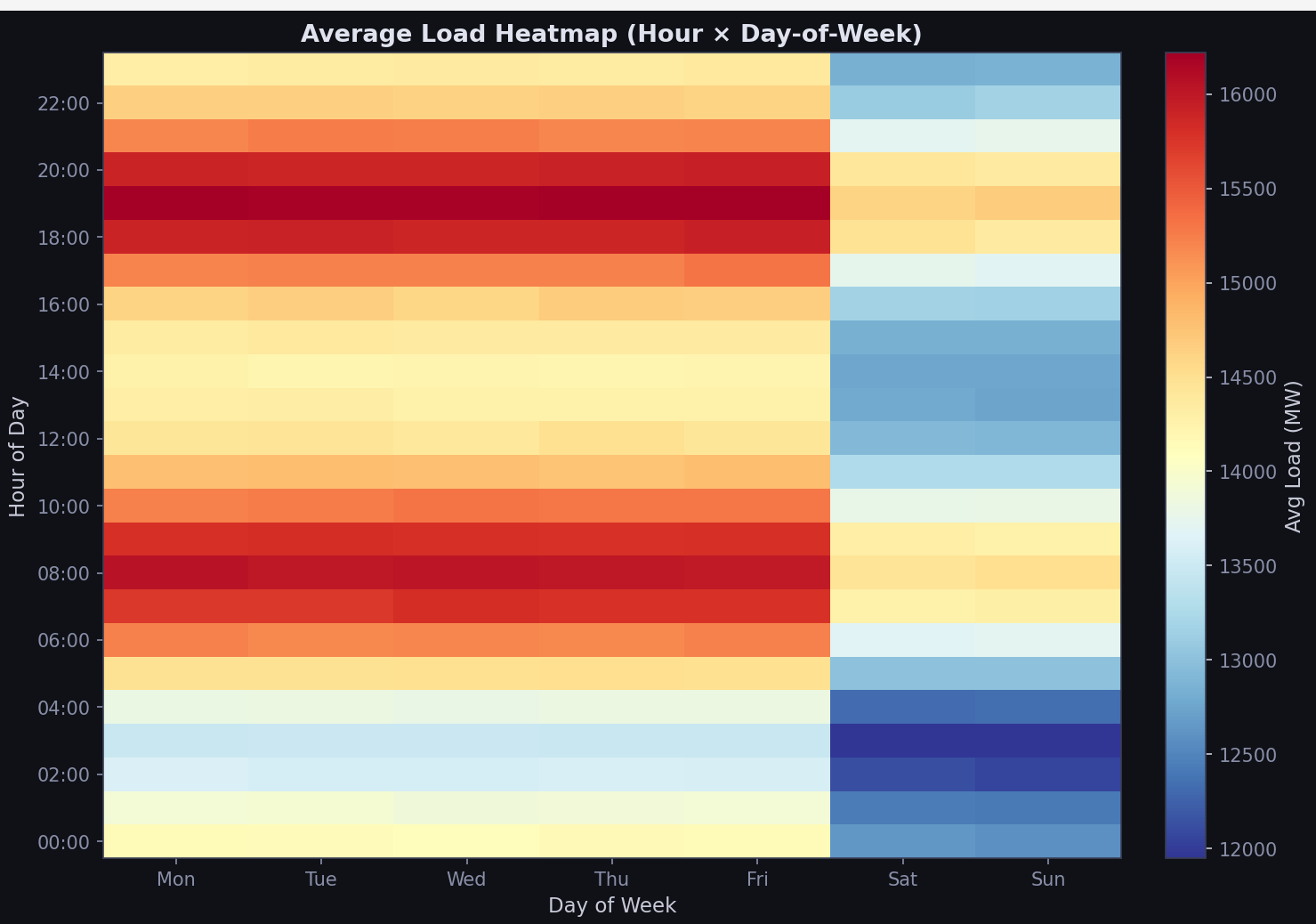
RQ2: Are GPU cluster ramp events detectable as distributional anomalies in regional load data?

EDA — Seasonality & Load Patterns



Full 7-year daily average, monthly distributions, and seasonal daily profiles. Summer mean load exceeds winter by 8.2%.

4. Temporal Heatmap (Hour × Day)



5. Dataset & Feature Engineering

PJM AEP Region — 61,321 hourly records, 2017–2023. PJM Interconnection, 13 U.S. states, ~65M customers. Load range: 9,000–24,000 MW. 183 injected anomalies.

12 engineered input features:

Category	Features
Calendar	hour, dayofweek, month, dayofyear, quarter, is_weekend
Lag (AR)	lag_1h, lag_2h, lag_24h, lag_168h
Rolling	roll_24h_mean, roll_24h_std

No data leakage: Last 6 months withheld as test set. All lag/rolling features computed causally on training data only.

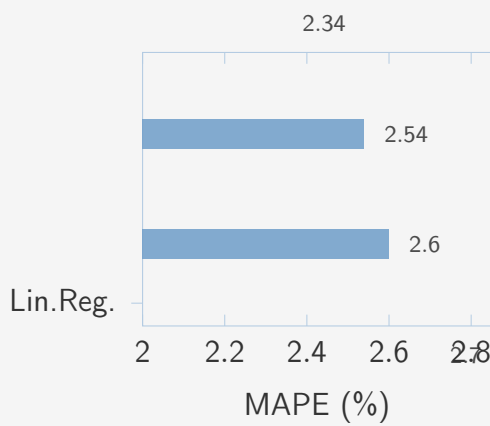
Forecast vs. Actual — First Test Week



White = actual; coloured dashes = models. XGBoost (green) tracks demand most closely. Lower panels: per-model residuals.

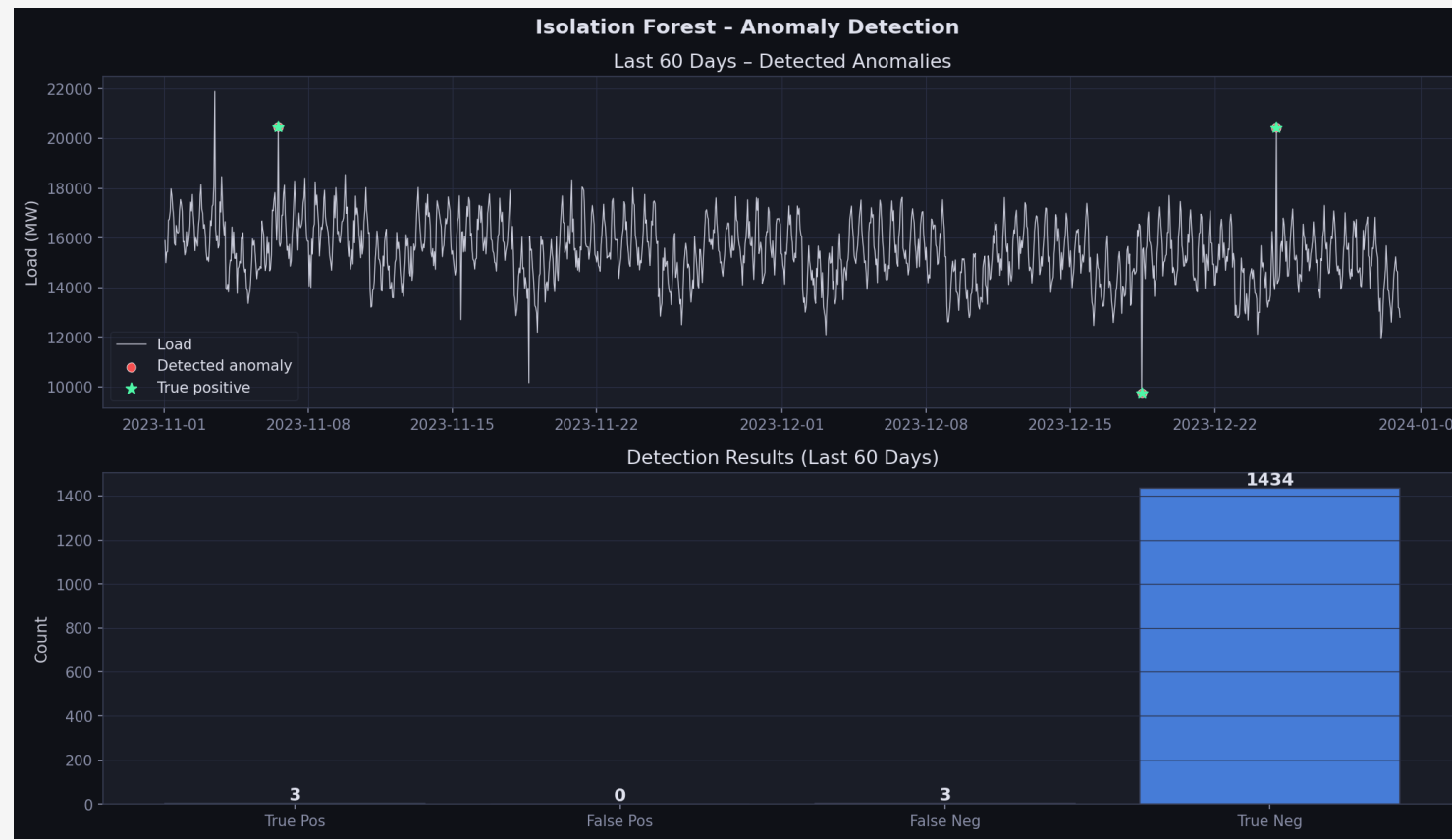
Model Performance Comparison

Model	MAE	RMSE	MAPE%
Lin. Reg.	437.6	607.8	2.70
Rand. F.	412.4	574.1	2.54
XGBoost	381.7	540.5	2.34
LSTM	≈420	≈580	≈2.60



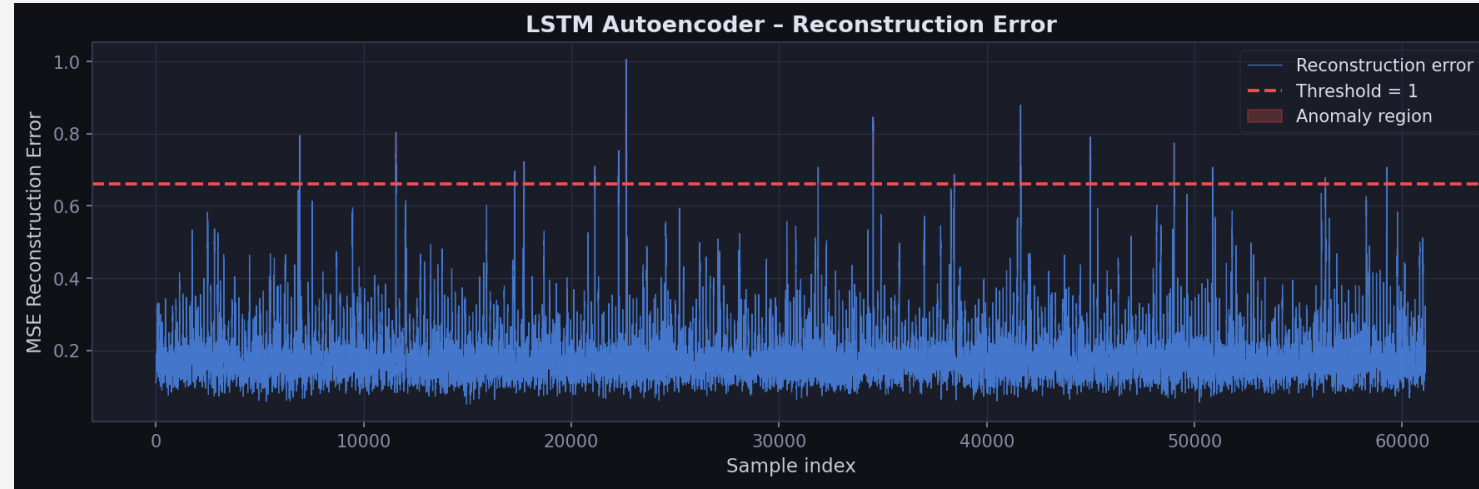
XGBoost achieves 2.34% MAPE — a 13.3% relative gain over the linear baseline. Gradient-boosted trees with lag features remain competitive vs. deep learning for univariate regional STLF.

Anomaly Detection — IF Results



Isolation Forest (100 trees, contamination=0.003). Red = detected; green * = true positives. F1 = 0.208.
Bar: TP/FP/FN/TN breakdown.

LSTM Autoencoder — Reconstruction Error



Red dashed = 3σ threshold (99.7th pct of training errors). Shaded regions = flagged anomaly windows. F1 = 0.006 — point anomalies diluted by window smoothing.

11. Anomaly Detection Performance

Model	Prec.	Recall	F1
Isolation Forest	0.207	0.209	0.208
LSTM Autoencoder	0.006	0.005	0.006

Recommendation: Hybrid deployment — Isolation Forest for point anomaly detection; LSTM Autoencoder for sustained behavioral drift (e.g. gradual data-center load growth). Each method targets a distinct anomaly regime.

Conclusions

- ✓ **XGBoost** achieves 2.34% MAPE on a strict temporal holdout — 13.3% relative improvement over linear baseline, surpassing LSTM
- ✓ **Isolation Forest** outperforms LSTM Autoencoder for unsupervised point anomaly detection (F1: 0.208 vs. 0.006) without any ground-truth labels
- ✓ **AI data center ramp events** are detectable as distributional anomalies, enabling proactive grid planning without labeled examples
- **Future work:** GNN multi-bus forecasting, Temporal Fusion Transformer, online SCADA anomaly detection, weather feature integration